

Noisy calibration, latent coverage and the location of the mean

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SUMMARY

Prediction intervals for a latent quantity, such as the mean of an unsampled area, are often calibrated on noisy proxies with Gaussian errors. A symmetric unimodal latent distribution makes such calibration conservative, but symmetry is seldom plausible. For bi-log-concave latent distributions, which include the log-concave ones and some multimodal ones, we show that the worst-case widening of the calibrated radius is governed by the distance between the latent mean and the boundary of the interval. At distance d it is at most the noise variance divided by $2d$, an order attained by asymmetric distributions with mean zero. Without a restriction on the mean it is at most c_q times the noise standard deviation, with a constant sharp for small noise, $c_{0.9} = 0.0191$, and that order requires the mean within a bounded multiple of the noise standard deviation of the boundary. A calibration slack of one part in a thousand at the ninety per cent level removes the widening, so the noisy conformal threshold at a suitable rank covers the latent target even when the noise variances are heterogeneous and unknown. Under log-concavity, known variances and a numerically computed radius shorten it by about ten per cent in our simulations.

Some key words: Conformal prediction; Deconvolution; Log-concavity; Measurement error; Small area estimation.

1. INTRODUCTION

Suppose that a residual W , the latent target minus a centre fitted on independent training data, is observed only through $V = W + e$, where $e \sim N(0, D)$ is independent of W and, until § 4.2, D is known. A survey statistician who calibrates a prediction interval for the true mean of an unsampled area on the direct estimates of sampled areas is in this position (Fay & Herriot, 1979), as is a conformal analyst whose calibration responses carry measurement error (Einbinder et al., 2024; Cohen et al., 2025). The threshold t at which the noisy scores $|V|$ have coverage p is observable. We ask which radius gives the latent residual coverage q , uniformly over a class of latent distributions.

If W is symmetric and unimodal, the theorem of Anderson (1955) gives $\text{pr}(|W + e| \leq t) \leq \text{pr}(|W| \leq t)$, so the noisy threshold is conservative (Einbinder et al., 2024). Without symmetry the available transfers make no shape assumption: a median-zero noise turns a noisy level $1 - \alpha$ into a latent level $1 - 2\alpha$ (Guille-Escuret & Ndiaye, 2024), and LatentCP (Zheng et al., 2026) uses a Markov-type compatibility threshold. Cohen et al. (2025) estimate the noise-free threshold by deconvolution, without a latent guarantee; conformal intervals for small areas (Bersson & Hoff, 2024) cover observable responses rather than latent area means; Armstrong et al. (2022) and Ignatiadis & Wager (2022) give robust and nonparametric empirical Bayes intervals for sampled units, and parametric small-area prediction intervals (Hall & Maiti, 2006; Yoshimori & Lahiri, 2014) rely on a model for the latent effects. For a fixed smooth latent distribution, Gaussian noise moves quantiles by $O(D)$ (Jochmans & Weidner, 2024).

Uniformly over a shape class the picture is different, and it is organized by the location of the mean. We work with log-concave laws and with the larger class of bi-log-concave laws (Dümbgen et al., 2017), which does not require unimodality. Without a restriction on the mean the worst-case widening at equal levels is at most $c_q D^{1/2}$, with a constant that is sharp as $D \rightarrow 0$ (Theorem 1); if the mean lies at distance d from the boundary of the interval the widening is at most $D/(2d)$ for every D , an order attained by asymmetric laws with mean zero (Theorem 2, Proposition 2); and a widening of order $D^{1/2}$ requires the mean within $O(D^{1/2})$ of the boundary, with a limiting coefficient given by a one-dimensional extremum (Theorem 3). The widening is also small: at $q = 0.9$ at most 1.16% of the radius for every bi-log-concave law, and numerically at most 0.22% for log-concave laws. A calibration slack of order 10^{-3} removes it (Corollary 1), so the noisy threshold at a suitable rank is valid for the latent target when the noise variances are heterogeneous and unknown (Proposition 4); known variances then serve efficiency (§ 4.2).

2. THE SHARP RADIUS

Let $\mathcal{L}$ be the class of log-concave laws on the real line, including point masses, and $\mathcal{B} \supset \mathcal{L}$ the class of laws that are point masses or have a continuous distribution function F with $\log F$ and $\log(1 - F)$ concave. Both classes are closed under translation and scaling, and convolution with a log-concave law preserves them (Prékopa, 1973; Saumard, 2019). After scaling t to 1, write $x = D/t^2$ and let $Z \sim N(0, 1)$ be independent of W . Define

$$R_{p,q}(x) = \sup\{Q_q(|W|) : W \in \mathcal{L}, \text{pr}(|W + x^{1/2}Z| \leq 1) \geq p\}, \quad (1)$$

where Q_q is the lower q -quantile, and $R_{p,q}(x) = -\infty$ if nothing is feasible, which happens exactly when $\text{pr}(|x^{1/2}Z| \leq 1) < p$. If calibration certifies noisy coverage at least p at threshold t , then $t R_{p,q}(D/t^2)$ is the smallest radius that guarantees latent coverage q for every log-concave latent law; this is sharpness within the class and for one constraint, not optimality among procedures that use the whole calibration sample. We call $R_{p,q}(x) - 1$ the radius correction; it is a correction to a length, not to a coverage probability. Let $R_{p,q}^{\mathcal{B}}$ be defined by (1) with $\mathcal{B}$ in place of $\mathcal{L}$, so $R_{p,q}^{\mathcal{B}} \geq R_{p,q}$. The upper bounds of §§ 3–4.1 hold for $R_{p,q}^{\mathcal{B}}$; the exact radius $R_{p,q}$ at a fixed x and its computation use log-concavity. For $0 \leq b \leq 1$ let $R_{p,q}^{(b)}(x)$ be defined by (1) with the additional constraint $|E(W)| \leq b$, so that $1 - b$ bounds the distance of the mean from the boundary.

Write $g_x(w) = \text{pr}(|w + x^{1/2}Z| \leq 1)$, so that the constraint in (1) reads $E\{g_x(W)\} \geq p$, and let $\mathcal{F}$ be the family of point masses and of laws with density proportional to $e^{\beta w}$ on a segment $[a, b]$.

PROPOSITION 1. $R_{p,q}(x) = \sup\{Q_q(|W|) : W \in \mathcal{F}, E\{g_x(W)\} \geq p\}$. The same holds for any continuous constraint kernel that vanishes at $\pm\infty$, in particular for the average kernel of § 4.2.

The proof, in the Supplementary Material, uses the one-constraint localization theorem of Fradelizi & Guédon (2004), whose extreme points are point masses and log-affine densities on segments; for these the problem reduces to a two-dimensional maximization of closed forms.

3. WORST-CASE RADIUS CORRECTION AND THE LOCATION OF THE MEAN

3.1. A one-sided problem

Near each end of the interval the problem is one-sided. For $u > 0$ let E_u be exponential with rate u , independent of Z , and for the exponential tail $b - E_u$ let

$$m_u(b) = \text{pr}(b - E_u + Z \leq 0) = \Phi(-b) + e^{-ub+u^2/2} \Phi(b - u),$$

which is continuous and strictly decreasing in b . Let $b_u(p)$ solve $m_u(b) = p$, and write $v_q(u) = b_u(q) - u^{-1} \log(1/q)$ and $k_q(u) = u^{-1} - b_u(q)$. The law $b_u(q) - E_u$ has q -quantile $v_q(u)$ and mean $-k_q(u)$. 80

LEMMA 1. Let $c_{p,q} = \sup\{F_Y^{-1}(q) : Y \in \mathcal{L}, \text{pr}(Y + Z \leq 0) \geq p\}$ and $c_q = c_{q,q}$. Then $c_{p,q} = \sup_{u>0}\{b_u(p) - u^{-1} \log(1/q)\}$, $c_q > 0$, and $q \mapsto c_q$ is nonincreasing. The same holds with $\mathcal{B}$ in place of $\mathcal{L}$.

LEMMA 2. Let $e^{-1} < q < 1$ and $\kappa \geq 0$. Then

$$\sup\{F_Y^{-1}(q) : Y \in \mathcal{B}, E(Y) \leq -\kappa, \text{pr}(Y + Z \leq 0) \geq q\} = L_q(\kappa) = \sup\{v_q(u) : u > 0, k_q(u) \geq \kappa\}.$$

The function L_q is positive and nonincreasing, and, for $q > 1/2$, $L_q(\kappa) = c_q$ for $\kappa \leq \kappa_q^*$, where κ_q^* is the largest value of k_q on the compact set of maximizers of v_q , and $\kappa L_q(\kappa) \rightarrow \{1 - \log(1/q)\}/2$ as $\kappa \rightarrow \infty$. 85

Exponential tails are extremal because concavity of $\log F_Y$ limits how fast mass can accumulate below the q -quantile, and they accumulate it as fast as allowed. Both proofs, in the Supplementary Material, replace Y by the exponential tail through a supergradient of $\log F_Y$ at its q -quantile; the replacement lies stochastically below Y , so it keeps the quantile, the noisy constraint and the bound on the mean. Ball arithmetic gives $c_{0.8} = 0.04607$, $c_{0.9} = 0.01906$ and $c_{0.95} = 0.00841$, and in double precision $\kappa_q^* = 6.81, 20.18$ and 50.07 . 90

3.2. No restriction on the mean

THEOREM 1. For every $q \in (0, 1)$, every $x > 0$ and every $W \in \mathcal{B}$ with $\text{pr}(|W + x^{1/2}Z| \leq 1) \geq q$, $Q_q(|W|) \leq 1 + c_q x^{1/2}$. Moreover $R_{q,q}(x) = 1 + c_q x^{1/2} + o(x^{1/2})$ as $x \rightarrow 0$, and likewise for $R_{q,q}^{\mathcal{B}}$. 95

Proof of the upper bound. Let $\alpha_R = \text{pr}(W + x^{1/2}Z > 1)$ and $\alpha_L = \text{pr}(W + x^{1/2}Z < -1)$, so that $\alpha_L + \alpha_R \leq 1 - q$. The law of $Y = (W - 1)/x^{1/2}$ has $\log F_Y$ concave and $\text{pr}(Y + Z \leq 0) = 1 - \alpha_R$. By Lemma 1, $F_Y^{-1}(1 - \alpha_R) \leq c_{1-\alpha_R} \leq c_q$, that is, $\text{pr}(W > 1 + c_q x^{1/2}) \leq \alpha_R$. The same argument applied to $-W$, which uses concavity of $\log(1 - F_W)$, gives $\text{pr}(W < -1 - c_q x^{1/2}) \leq \alpha_L$, so $\text{pr}(|W| \leq 1 + c_q x^{1/2}) \geq q$. 100 $\square$

The lower bound, proved in the Supplementary Material, uses $W = 1 + x^{1/2}\{b_u(q + \eta_x) - E_u\}$, whose density rises to a cut-off just outside the threshold and whose mean is within $O(x^{1/2})$ of the boundary. Since $E\{g_x(W)\} \leq g_x(0)$, the constraint is feasible only if $x^{1/2} \leq 1/\Phi^{-1}\{(1+q)/2\}$, so Theorem 1 gives $Q_q(|W|) - 1 \leq c_q/\Phi^{-1}\{(1+q)/2\}$, which is below 1.16% at $q = 0.9$ for every $W \in \mathcal{B}$. For log-concave laws, double-precision upper bounds give $R_{q,q}(x) \leq 1$ for $x \geq 0.013$, hence a correction below 0.22%, and maximization over $\mathcal{F}$ finds 0.063% (Supplementary Material). 105

3.3. The mean away from the boundary

THEOREM 2. Let $W \in \mathcal{B}$ with $|E(W)| \leq B$, let $\sigma > 0$ and $t > B$, and suppose that $\text{pr}(|W + \sigma Z| \leq t) \geq q$. Then 110

$$Q_q(|W|) \leq B + \{(t - B)^2 + \sigma^2\}^{1/2} \leq t + \frac{\sigma^2}{2(t - B)}.$$

In particular $R_{p,q}^{(b)}(x) \leq 1 + x/\{2(1 - b)\}$ for all $p \geq q$ and $x > 0$, and $Q_q(|W|) \leq (t^2 + \sigma^2)^{1/2}$ if $E(W) = 0$.

A widening needs a density that rises steeply towards an end of the interval, and a mean far from that end caps the rate of rise. The proof in the Appendix makes this precise with an 115

exponential-tail comparison and the heat equation for quantiles (Steinbrecher & Shaw, 2008), which give $d\{r(s) - B\}^2/ds \geq -1$ for $r(s) = Q_q(|W + s^{1/2}Z|)$; the bound is uniform over the class and needs neither unimodality nor symmetry. It improves on Theorem 1 only when $x^{1/2} < 2c_q(1 - b)/(1 - c_q^2)$; for mean-zero laws at $q = 0.9$ this means $x < 1.5 \times 10^{-3}$, where both bounds are below 0.08%, so its role is to explain when a widening can arise rather than to change a radius. The order x cannot be improved, even for mean-zero laws.

PROPOSITION 2. *Let X have density proportional to $e^{\beta X}$ on $[-\ell, 0]$ with $\beta > 0$ and $0 < \ell \leq \infty$, let $W = X - E(X)$ and let $r_0 = Q_q(|W|)$. If $\pm r_0$ lie in the interior of the support of W , then, after scaling W so that the noisy q -quantile of $|W + x^{1/2}Z|$ equals 1,*

$$Q_q(|W|) = 1 + \frac{1}{2}\beta r_0 \tanh(\beta r_0) x + o(x) \quad (x \rightarrow 0).$$

For every $q \in (0, 1)$ the condition holds for small $\beta\ell$, so $R_{q,q}^{(0)}(x) - 1$ is of exact order x .

Numerically the best coefficient in this family is 0.3466 at $q = 0.8$, for $W = 1 - E_1$, and 0.0747 at $q = 0.9$, so the worst case over centred laws lies in $[0.346, 0.5]$ and $[0.0747, 0.5]$.

3.4. The transition

THEOREM 3. *Let $1 - e^{-1} < q < 1$, $x > 0$ and $W \in \mathcal{B}$ with $\text{pr}(|W + x^{1/2}Z| \leq 1) \geq q$. Then $|E(W)| < 1$, and if $\delta = Q_q(|W|) - 1 > 0$,*

$$2\{1 - |E(W)|\}\delta + \delta^2 \leq x. \quad (2)$$

Moreover, for every $\kappa \geq 0$,

$$\lim_{x \rightarrow 0} x^{-1/2} \{R_{q,q}^{(b_x)}(x) - 1\} = L_q(\kappa), \quad b_x = 1 - \kappa x^{1/2},$$

and the same limit holds with $\mathcal{L}$ replaced by $\mathcal{B}$ in the definition of $R^{(b)}$.

By (2), a correction $\delta \geq ax^{1/2}$ with $0 < a < 1$ forces $1 - |E(W)| \leq (1 - a^2)x^{1/2}/(2a)$. In the original units the mean lies $\kappa D^{1/2}$ from the threshold, and the unrestricted worst case $c_q D^{1/2}$ persists until κ exceeds κ_q^* , about 20 noise standard deviations at $q = 0.9$; beyond it the coefficient decays like $\{1 - \log(1/q)\}/(2\kappa)$ (Fig. 1). The first part is proved in the Appendix and the limit in the Supplementary Material.

4. CALIBRATION SLACK AND A FINITE-SAMPLE PROCEDURE

4.1. Slack removes the correction

The proof of Theorem 1 splits the noisy failure budget between the two ends. With a noisy level $p \geq q$, splitting both the noisy and the latent budgets gives, for every x and every $W \in \mathcal{B}$ feasible at level p , $Q_q(|W|) \leq 1 + C_{p,q}x^{1/2}$, where $C_{p,q} \geq c_{p,q}$ is a worst case over splits of the one-sided constants, defined in the Supplementary Material. A slack $p - q$ lets the noisy constraint absorb the one-sided widening at both ends, and once $C_{p,q} \leq 0$ no widening is needed at any noise level.

COROLLARY 1. *If $C_{p,q} \leq 0$ then $R_{p,q}^{\mathcal{B}}(x) \leq 1$ for every x , and if $c_{p,q} > 0$ then $R_{p,q}(x) > 1$ for all small x . The smallest slack $p - q$ with $C_{p,q} \leq 0$ lies in $(4.3 \times 10^{-3}, 5.0 \times 10^{-3}]$, $(7.9 \times 10^{-4}, 1.0 \times 10^{-3}]$ and $(1.5 \times 10^{-4}, 2.0 \times 10^{-4}]$ at $q = 0.8, 0.9$ and 0.95 , and so does the smallest slack with $R_{q+\delta,q}(x) \leq 1$ for all x .*

Ball arithmetic also gives $C_{0.9068,0.9} \leq -0.114$, used below.

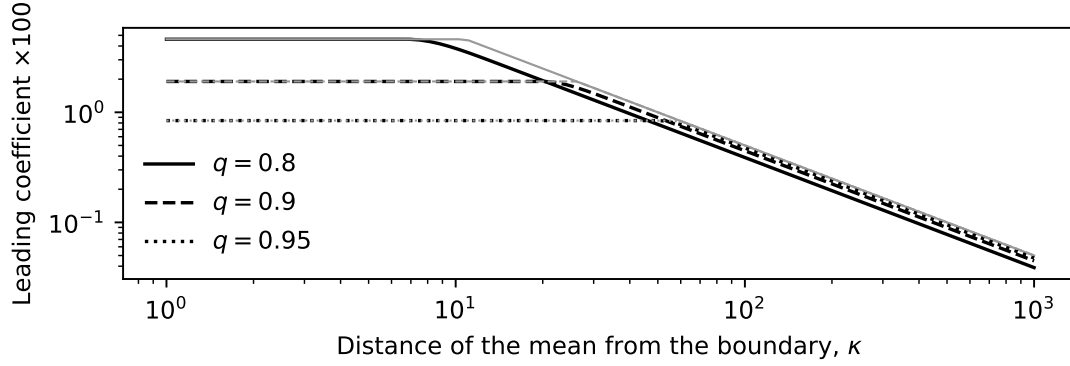


Fig. 1. Leading coefficient $L_q(\kappa)$ of the worst-case radius correction, times 100, when the mean lies at distance at least $\kappa x^{1/2}$ from the boundary (Theorem 3), for $q = 0.8$ (solid), 0.9 (dashes) and 0.95 (dots), with the upper bound $\min\{c_q, (\kappa^2 + 1)^{1/2} - \kappa\}$ from Theorems 1 and 2 (grey). Both axes are logarithmic.

Alt text: Line plot on logarithmic axes of the leading coefficient against the scaled distance of the mean from the boundary for three coverage levels. Each curve is flat at its unrestricted constant up to a threshold near 7, 20 and 50 and then decreases in proportion to the reciprocal of the distance, slightly below a grey upper bound.

4.2. Heterogeneous and unknown noise variances

Let $W_1, \dots, W_K, W_{\text{new}}$ be independent with law G , independent of $e_i \sim N(0, D_i)$ with the D_i fixed. The scores $V_i = W_i + e_i$ are independent but not identically distributed, and $\mathcal{D}$ denotes the calibration data.

PROPOSITION 3. *Let $T = |V|_{(k)}$, let $\bar{F}$ be the average distribution function of the $|V_i|$, assumed continuous, and let p_k be the δ -quantile of the Beta($k, K + 1 - k$) distribution. If $k \geq Kp_k + 1$ then $\text{pr}\{\bar{F}(T) \geq p_k\} \geq 1 - \delta$.*

The proof, by Hoeffding (1956), is in the Supplementary Material; the result is a case of the comparison of Sen (1970).

PROPOSITION 4. *Let $G \in \mathcal{B}$, let the D_i be unknown, and suppose that $k \geq Kp_k + 1$ and $C_{p_k, q} \leq 0$. Then $\text{pr}_{\mathcal{D}}\{\text{pr}(|W_{\text{new}}| \leq T \mid \mathcal{D}) \geq q\} \geq 1 - \delta$.*

The slack absorbs the worst widening for every noise variance at once, so averaging over unknown variances cannot break it.

Proof. On the event of Proposition 3, $K^{-1} \sum_i \text{pr}(|W + e_i| \leq T) \geq p_k$ for $W \sim G$. If $\text{pr}(|W| \leq T) < q$, Corollary 1 at threshold T gives $\text{pr}(|W + e_i| \leq T) < p_k$ for every i , whatever D_i , and so for the average, a contradiction. $\square$

At $q = 0.9$ and $\delta = 0.05$ the conditions hold at the smallest rank with $p_k \geq 0.901$ and $k \geq Kp_k + 1$, which exists for $K \geq 29$ and, for $K \leq 3000$, is at most three ranks above the usual rank with $p_k \geq 0.9$ (Supplementary Material). For $K = 110$ the two coincide, at $k = 105$ with $p_k = 0.9068$. Known variances shorten the threshold.

COROLLARY 2. *Under the conditions of Proposition 4, if $D_i \geq D_{\min}$ for all i and $C_{p', q} < 0$ for some $p' \leq p_k$, then $s = \max\{0, T + C_{p', q} D_{\min}^{1/2}\}$ has the same guarantee.*

Proof. On the event of Proposition 3, some i has $\text{pr}(|W + e_i| \leq T) \geq p_k \geq p'$, and the bound of § 4.1 at $x = D_i/T^2$ gives $Q_q(|W|) \leq T + C_{p', q} D_i^{1/2} \leq T + C_{p', q} D_{\min}^{1/2}$. $\square$

Table 1. *Reliability, times 100, and mean width, in parentheses; target reliability 95*

latent law	Markov	Fay–Herriot	noisy threshold	Corollary 2	computed
normal	100 (5.92)	96 (3.93)	100 (4.96)	100 (4.90)	98 (4.45)
Laplace	100 (6.29)	93 (3.96)	100 (5.15)	100 (5.09)	99 (4.66)
centred gamma	100 (6.16)	100 (3.95)	100 (4.97)	100 (4.90)	100 (4.45)
truncated Laplace	100 (5.89)	85 (3.89)	100 (4.93)	100 (4.87)	97 (4.42)
truncated exponential	100 (6.17)	95 (3.93)	100 (5.06)	100 (4.99)	100 (4.55)
bimodal ($\mathcal{B}$ only)	100 (5.77)	100 (4.01)	100 (4.87)	100 (4.81)	99 (4.35)
t_3 (outside $\mathcal{B}$)	100 (5.70)	97 (3.80)	100 (4.66)	100 (4.60)	98 (4.09)

Noisy threshold, T at rank 105; computed, TR^{mix} ; centred gamma with shape 2; truncated exponential, standardized density proportional to e^{4w} on $[0, 1]$; bimodal, $0.4 N(-0.9487, 0.6325^2) + 0.6 N(0.6325, 0.6325^2)$; widths are full widths. The Markov rule, which needs no shape assumption, is guaranteed for every law. Under the bimodal law the noisy threshold and Corollary 2 are guaranteed and the computed radius is exploratory; under t_3 the noisy threshold is guaranteed by Anderson’s theorem, since the law is symmetric and unimodal, while Corollary 2 and the computed radius are not. The binomial standard error of a reliability is 1.8 at 95 and 2.9 at the lowest reliability shown; standard errors of widths are at most 0.13.

At $q = 0.9$ this gives $s = T - 0.114D_{\min}^{1/2}$; plugging in the smallest variance instead is not valid (Supplementary Material). Among rules that use the single constraint at T , the sharpest use of known variances is the average kernel $\bar{g}(w) = K^{-1} \sum_i \text{pr}(|w + e_i| \leq T)$: $\int \bar{g} dG \geq p_k$ on the event of Proposition 3, Proposition 1 applies for log-concave G , and the computed radius $s = TR_{p_k - \varepsilon, q}^{\text{mix}}(D_1/T^2, \dots, D_K/T^2)$, with ε bounding a discretization error, has the same guarantee.

Proposition 4 and Corollary 2 rest on proofs and on constants certified in ball arithmetic. The computed radius with the exact R^{mix} is valid for log-concave G ; our implementation returns a numerical upper bound on R^{mix} , from a branch and bound with double-precision closed forms and a margin of 10^{-9} , not an interval-arithmetic certificate.

5. NUMERICAL ILLUSTRATION

The target is $\text{pr}_{\mathcal{D}}\{\text{pr}(W_{\text{new}} \in C \mid \mathcal{D}) \geq 0.9\} \geq 0.95$, with $K = 110$, $\text{var}(W) = 1$ and heterogeneous noise variances $D_i = 0.577 L_i / E(L_i)$, L_i lognormal with log-scale 0.7. Latent residuals, all with mean zero, are drawn directly; conditional coverage given each data set is exact, and reliability is estimated from 150 data sets per law. Besides four log-concave laws we use a truncated exponential law in the extremal family $\mathcal{F}$, an asymmetric bimodal mixture in $\mathcal{B}$ but not in $\mathcal{L}$, and a t_3 law, unimodal but outside $\mathcal{B}$. We compare the three rules of § 4.2; a Fay–Herriot tolerance interval, our adaptation of the parametric bootstrap of Chatterjee et al. (2008) (Supplementary Material); and the Markov rule applied to $\bar{g}_T(0) - \bar{g}_T(W) \geq 0$, the smallest radius the single constraint allows without a shape assumption, at rank 107, the best of 106–110 on average over the first four laws.

The rules with a guarantee reached the target for every law in their class, and the computed radius did so also under the bimodal and t_3 laws, for which it has none. Corollary 2 was 1.2–1.4% shorter than the noisy threshold, and the computed radius 9.5–12% shorter than the noisy threshold and 25–28% shorter than the Markov rule. The Fay–Herriot interval was 7–15% shorter than the computed radius but missed the target under the truncated Laplace law. With 2000 data sets per law the noisy threshold, Corollary 2 and the Markov rule reached conditional coverage 0.9 in every data set, with mean conditional coverage 0.97–0.99 for the first two, so the guarantees are conservative for laws far from the extremal ones of § 3. The Supplementary

Material compares LatentCP (Zheng et al., 2026), 16–18% longer than the computed radius, and the deconvolution threshold of Cohen et al. (2025), which has no latent guarantee, and shows that at an oracle-matched reliability the gains lie in the cost of a guarantee rather than in the shape of the interval. 205

6. DISCUSSION

The guarantees are conditional on the calibration sample and concern a new area whose residual is drawn from the law of the calibration residuals, not areas that differ systematically from the sampled ones; they assume independent Gaussian noise and a bi-log-concave, or for the computed radius log-concave, latent law. The mean condition of Theorems 2 and 3 concerns the latent residual law, which an intercept in the fitted regression does not centre. 210

Open questions include robustness to non-Gaussian noise, the leading coefficient for centred laws, between 0.0747 and 0.5 at $q = 0.9$, a median version of Theorem 2, which Gaussian smoothing does not preserve, a proof that $C_{p,q} = c_{p,q}$, and whether $R_{p,q}^{\mathcal{B}} = R_{p,q}$ at fixed x . 215

DECLARATION OF THE USE OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES

[To be completed by the author.]

SUPPLEMENTARY MATERIAL

The Supplementary Material includes omitted proofs, the certificates, the estimated-variance procedure, further simulations and an application. Code reproducing every number is at <https://github.com/kunhailP/noisy-calibration-latent-coverage>. 220

APPENDIX 1

Proofs for § 3

Proof of Theorem 2. Let $\mu = E(W)$. If W is a point mass, $Q_q(|W|) = |\mu| \leq B$. Otherwise, for $s > 0$ the law of $W + s^{1/2}Z$ is in $\mathcal{B}$ (Saumard, 2019), with mean μ , distribution function F_s and a positive smooth density f_s . For $r > \mu$ let $h = f_s(r)/F_s(r)$. The tangent of $\log F_s$ at r gives $F_s(y) \leq \min\{F_s(r)e^{h(y-r)}, 1\}$, the distribution function of an exponential tail with mean $r - \{1 + \log F_s(r)\}/h$ that lies stochastically below $W + s^{1/2}Z$, so $\mu \geq r - \{1 + \log F_s(r)\}/h$ and $h(r - \mu) \leq 1$. Laws in $\mathcal{B}$ satisfy $f'_s \leq f_s^2/F_s$ (Dümbgen et al., 2017), so $(\log f_s)'(r) \leq h \leq 1/(r - \mu)$; applied to $-W$ this gives $(\log f_s)'(-r) \geq -1/(r + \mu)$. Hence, for $r > B$, 225

$$f'_s(r) \leq \frac{f_s(r)}{r - B}, \quad -f'_s(-r) \leq \frac{f_s(-r)}{r - B}.$$

Let $H_s(r) = \text{pr}(|W + s^{1/2}Z| \leq r)$. By the heat equation $\partial_s f_s = f_s''/2$, 230

$$\partial_s H_s(r) = \frac{1}{2} \{f'_s(r) - f'_s(-r)\} \leq \frac{f_s(r) + f_s(-r)}{2(r - B)} = \frac{\partial_r H_s(r)}{2(r - B)}.$$

The quantile $r(s) = Q_q(|W + s^{1/2}Z|)$ solves $H_s\{r(s)\} = q$ and is continuously differentiable on $s > 0$, with $r'(s) \geq -1/[2\{r(s) - B\}]$ where $r(s) > B$. Hence $\{r(s) - B\}_+^2$ has derivative at least -1 almost everywhere, and $\{r(\epsilon) - B\}_+^2 \leq \{r(\sigma^2) - B\}_+^2 + \sigma^2 - \epsilon$. As $\epsilon \rightarrow 0$, $r(\epsilon) \rightarrow Q_q(|W|)$, because a nondegenerate law in $\mathcal{B}$ has a positive density on the interior of its support (Dümbgen et al., 2017), so the distribution function of $|W|$ is continuous and strictly increasing where it lies strictly between 0 and 1. Since $r(\sigma^2) \leq t$, the first inequality follows, and the second from $(d^2 + \sigma^2)^{1/2} \leq d + \sigma^2/(2d)$. 235 $\square$

Proof of Theorem 3, first part. Let $V = W + x^{1/2}Z$ with distribution function F . Since $\log F$ and $\log(1 - F)$ are concave and $F(V)$ is uniform, $E\{\log F(V)\} = -1$ and Jensen's inequality gives $F\{E(V)\} \geq e^{-1}$ and likewise $1 - F\{E(V)\} \geq e^{-1}$. If $E(W) \geq 1$ then $\text{pr}(|V| \leq 1) \leq F\{E(W)\} \leq 1 - e^{-1} < q$, and similarly if $E(W) \leq -1$; so $|E(W)| < 1$. Theorem 2 with $B = |E(W)|$ and $t = 1$ gives $\delta \leq \{d^2 + x\}^{1/2} - d$ with $d = 1 - |E(W)|$, which is (2). $\square$

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